

A black caterpillar with orange spots and spines is crawling on a green leaf. The caterpillar has a segmented body with several pairs of sharp, reddish-brown spines. It is positioned horizontally across the frame, with its head on the left and tail on the right. The background is a blurred green leaf.

Behavioral plasticity

IB 431 – Behavioral Ecology
Fall 2025
September 15th 2025

Outline

- I. Phenotypic plasticity and GxE
- II. Behavioral plasticity and behavioral reaction norms
- III. Evolution of plasticity
- IV. Go over Nielsen's hypotheses
- V. Discuss Nielsen et al 2018

Phenotypic plasticity

- Expression of phenotype depends on environment

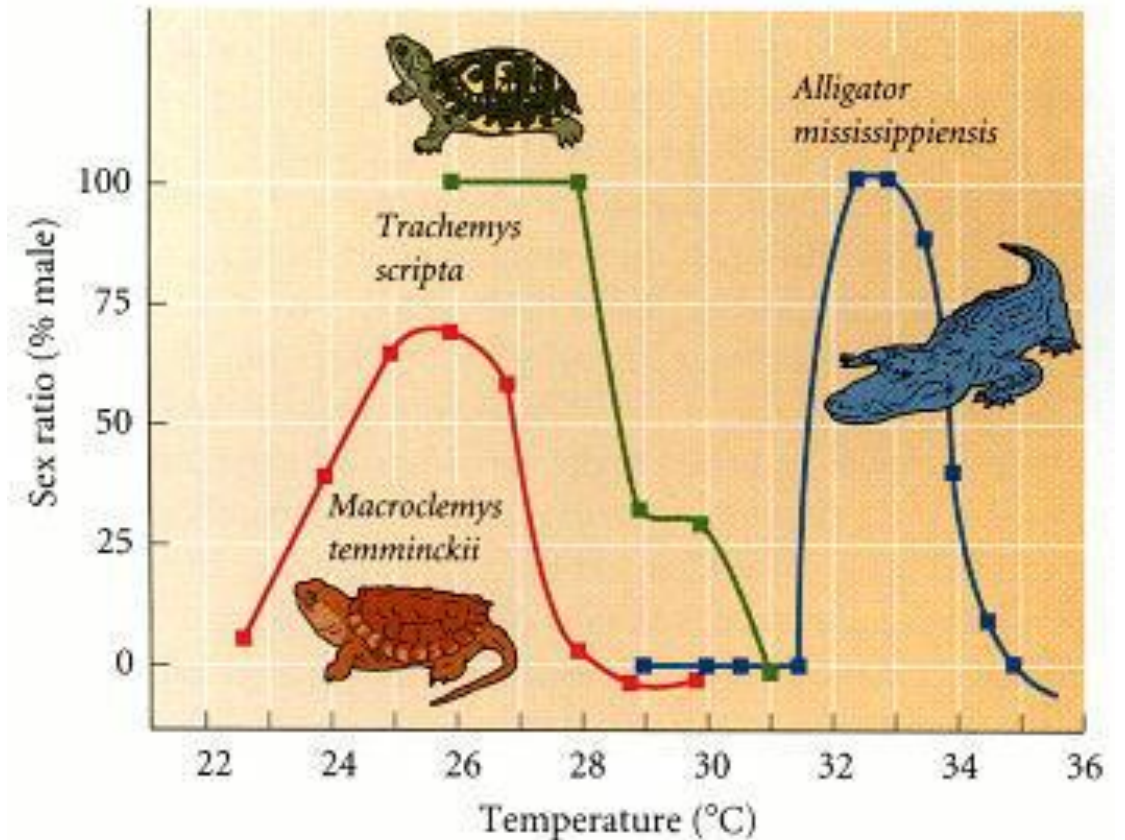
Phenotypic plasticity

- Expression of phenotype depends on environment
- ***Daphnia* grow protective helmet when exposed to cues of predators**



Phenotypic plasticity

- Expression of phenotype depends on environment
- *Daphnia* grow protective helmet when exposed to cues of predators
- **Temperature-dependent sex determination in reptiles**

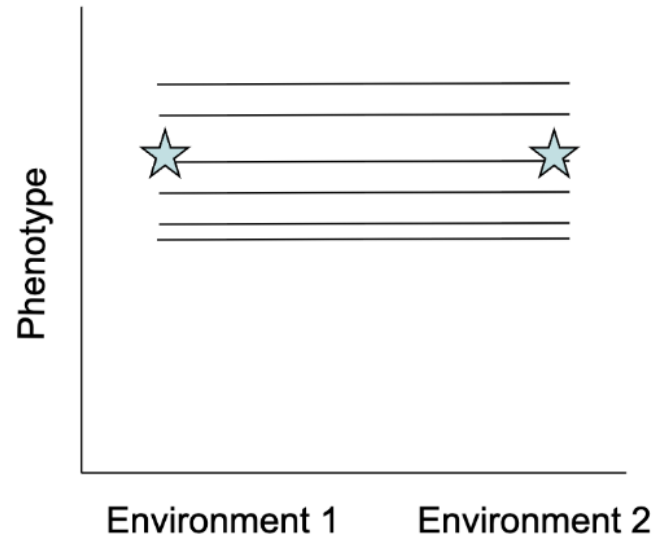


Reaction norm

Range of phenotypes
that can be produced
by a given genotype
across environments

Reaction norm

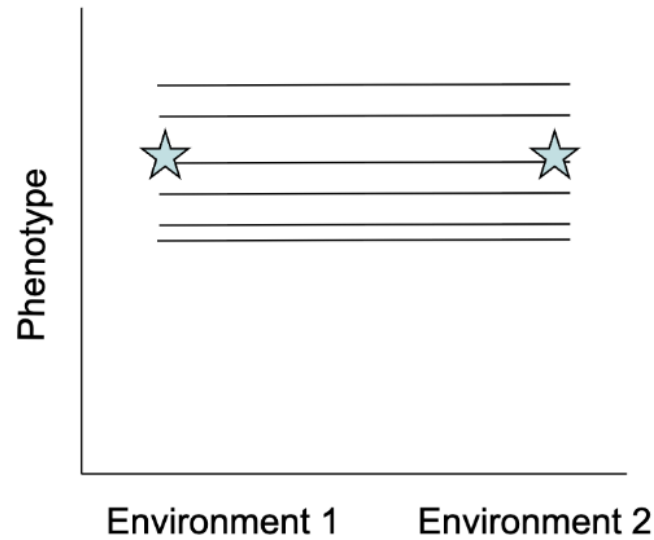
Range of phenotypes that can be produced by a given genotype across environments



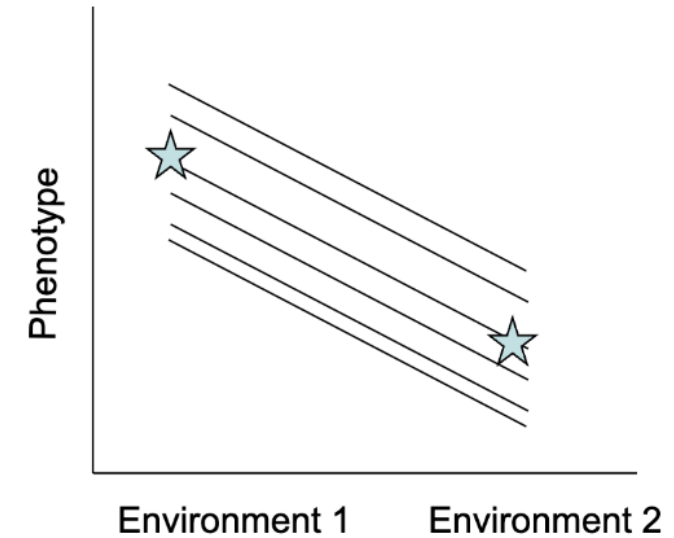
Effect of genotype,
no environmental effect

Reaction norm

Range of phenotypes that can be produced by a given genotype across environments



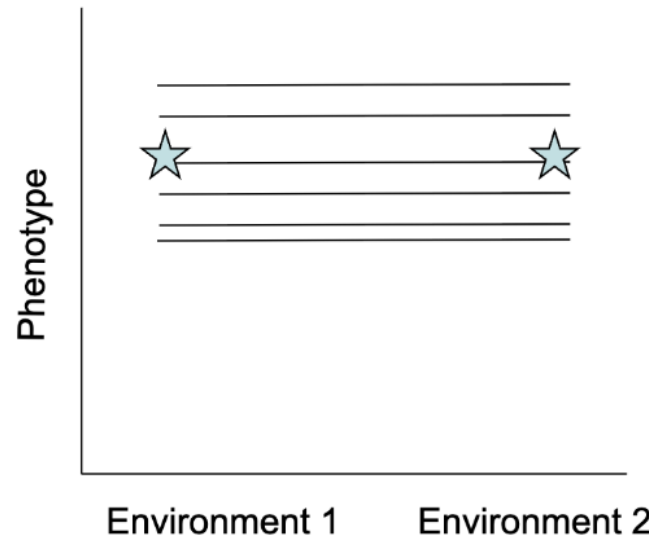
Effect of genotype,
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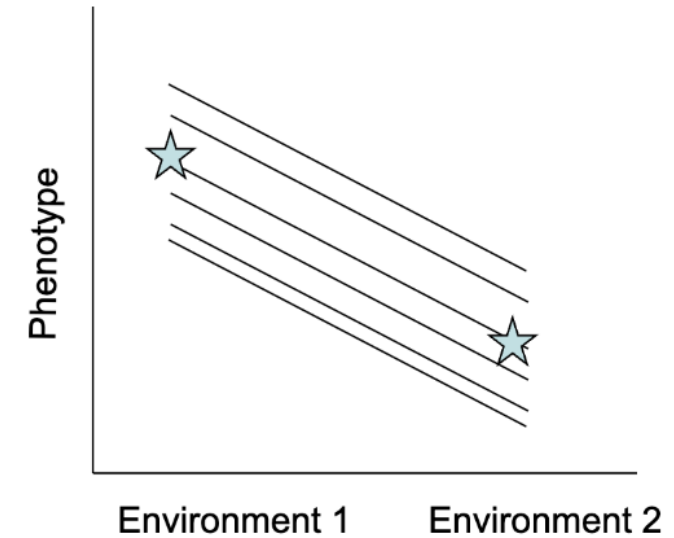
Effect of genotype and
environment

Reaction norm

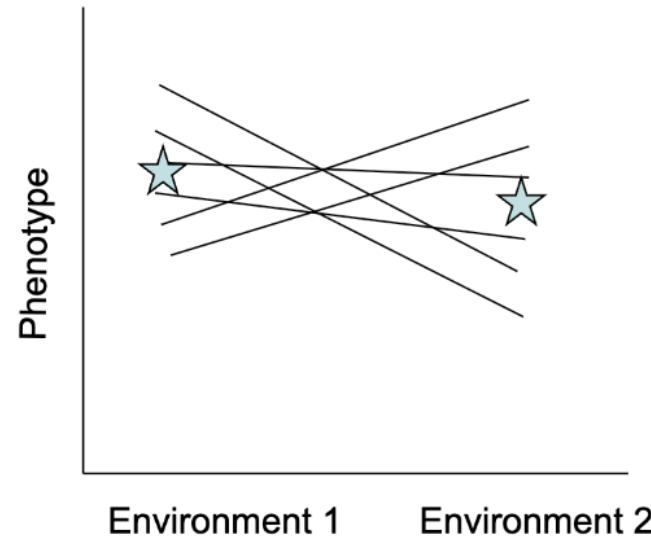
Range of phenotypes that can be produced by a given genotype across environments



Effect of genotype,
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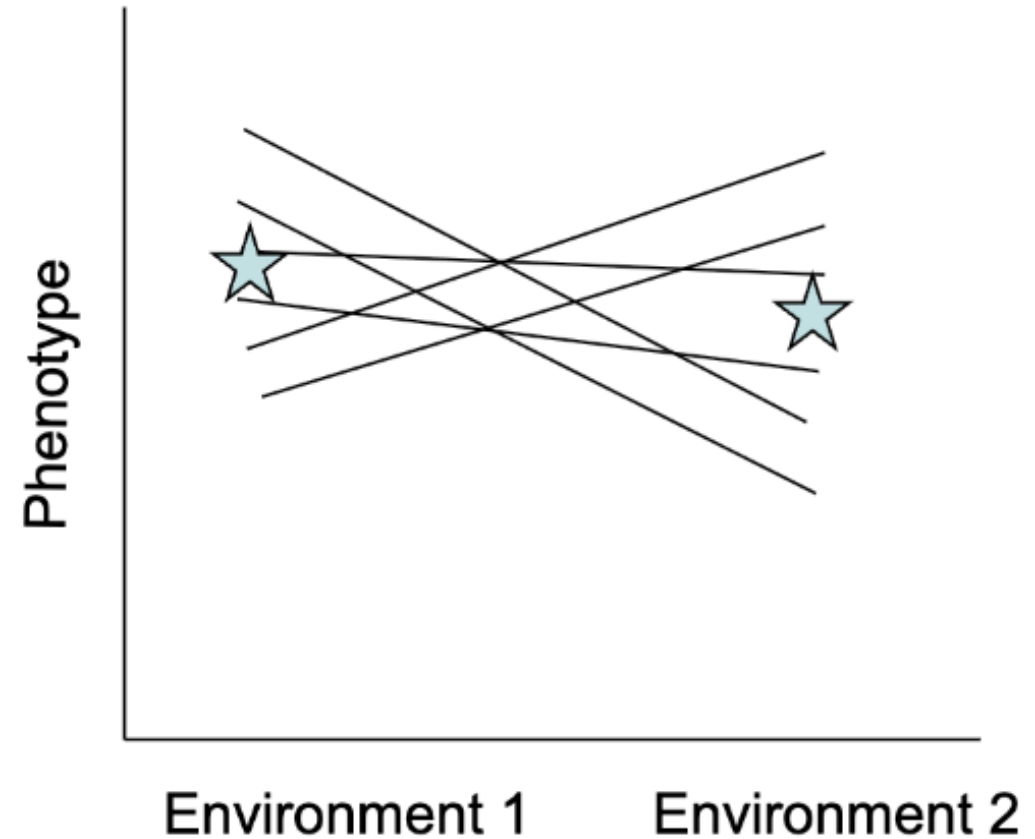
Effect of genotype and
environment



Genotype by environment
interaction (GxE)

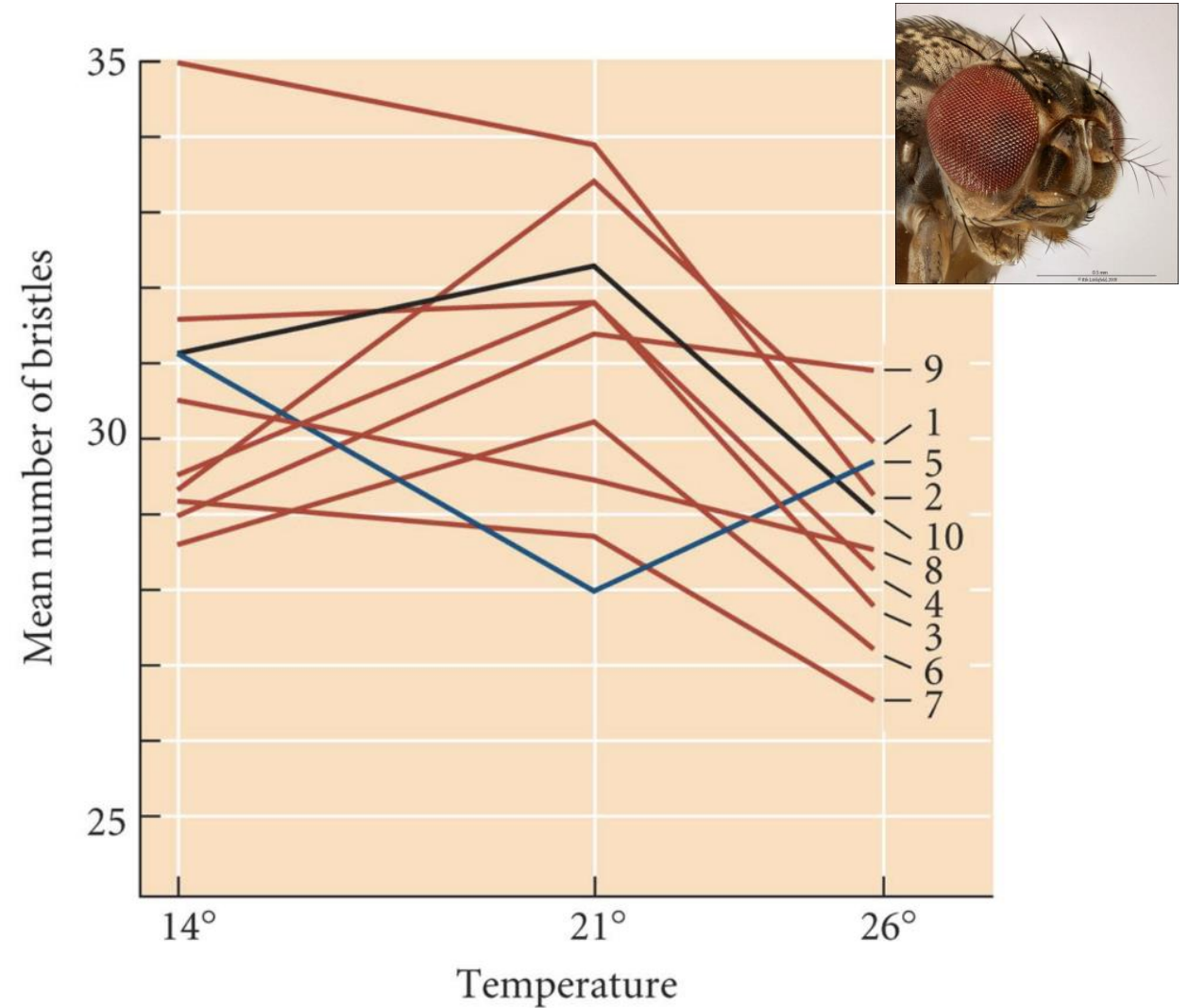
GxE interactions

- Effect of the environment on phenotype depends on the genotype



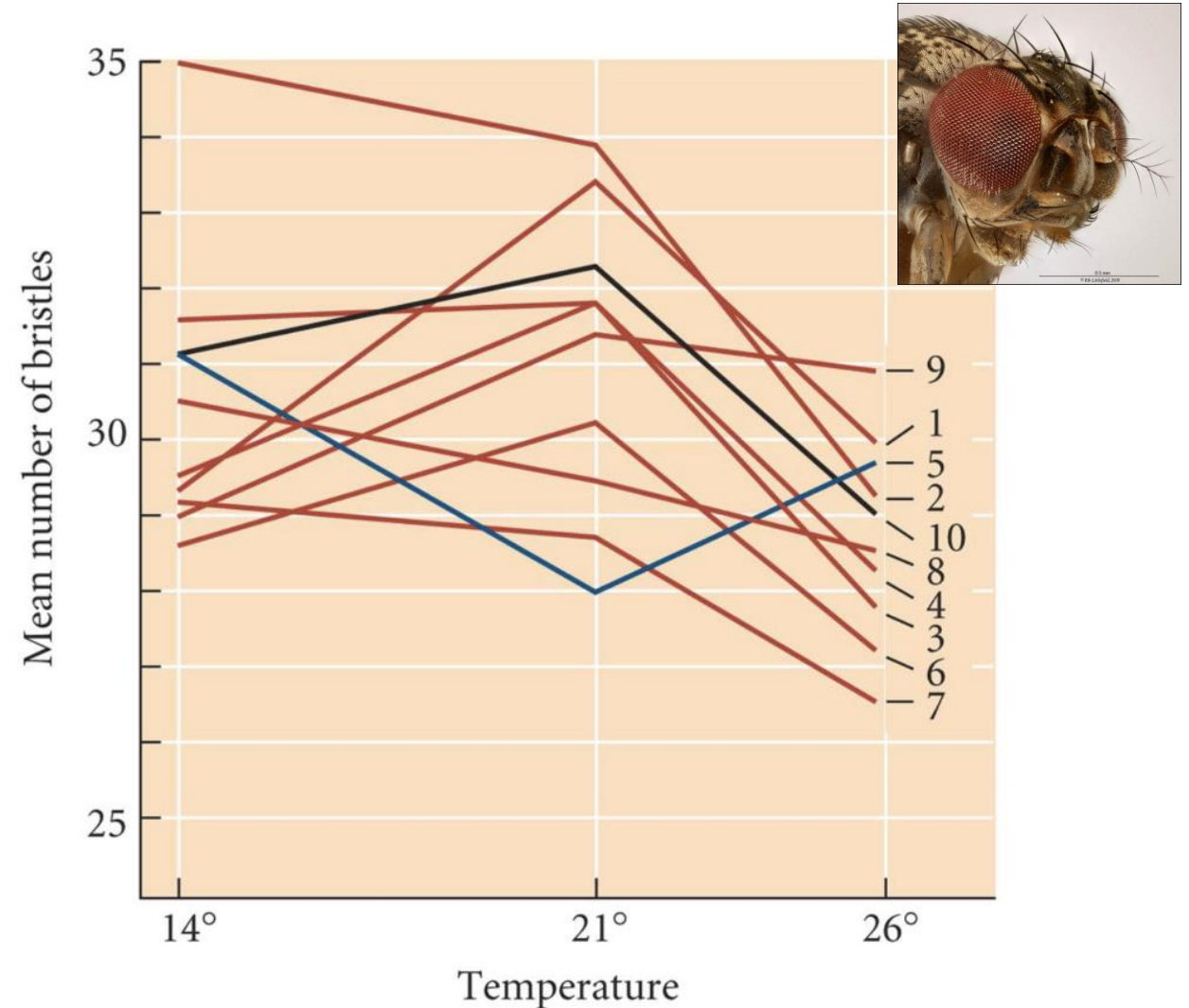
GxE interactions

- GxE for number of bristles produced by fruit flies



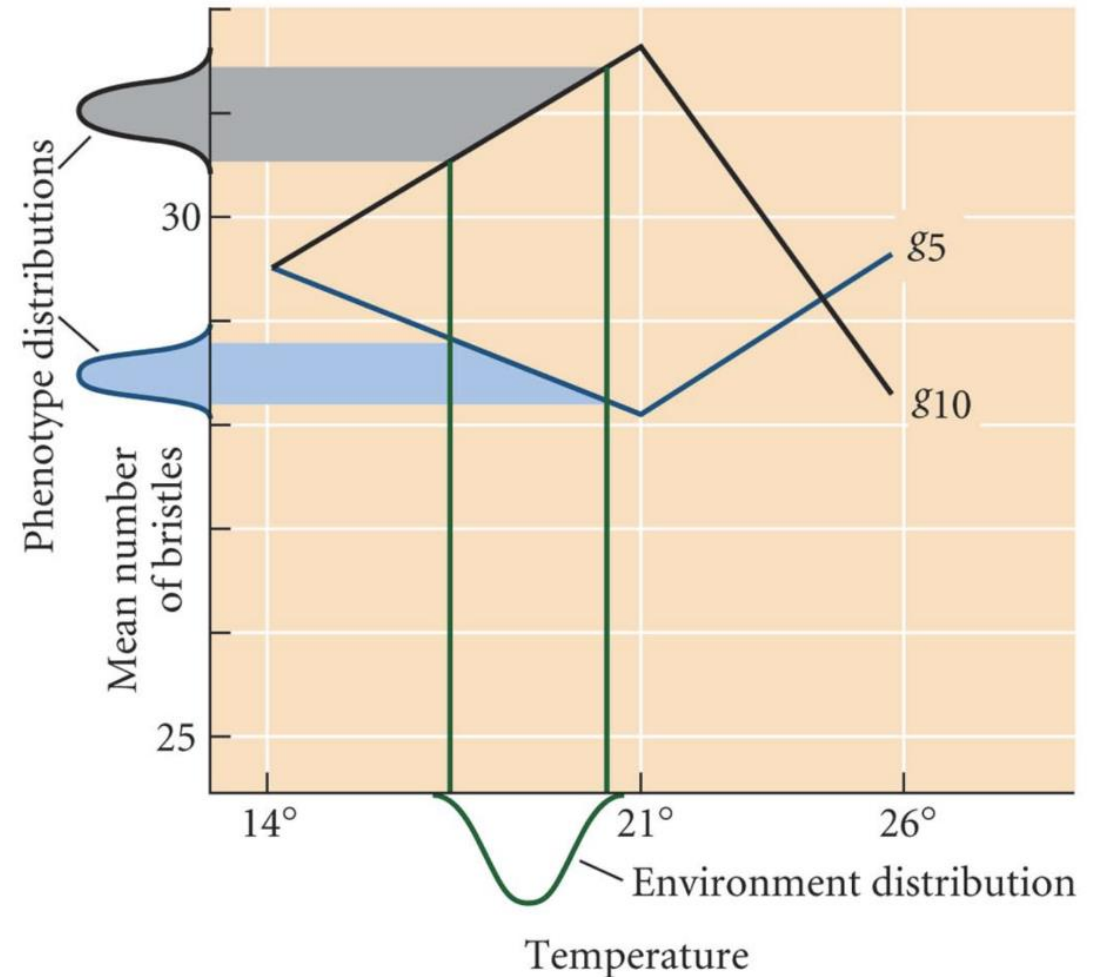
GxE interactions

- GxE for number of bristles produced by fruit flies
- Not all genotypes are equally plastic

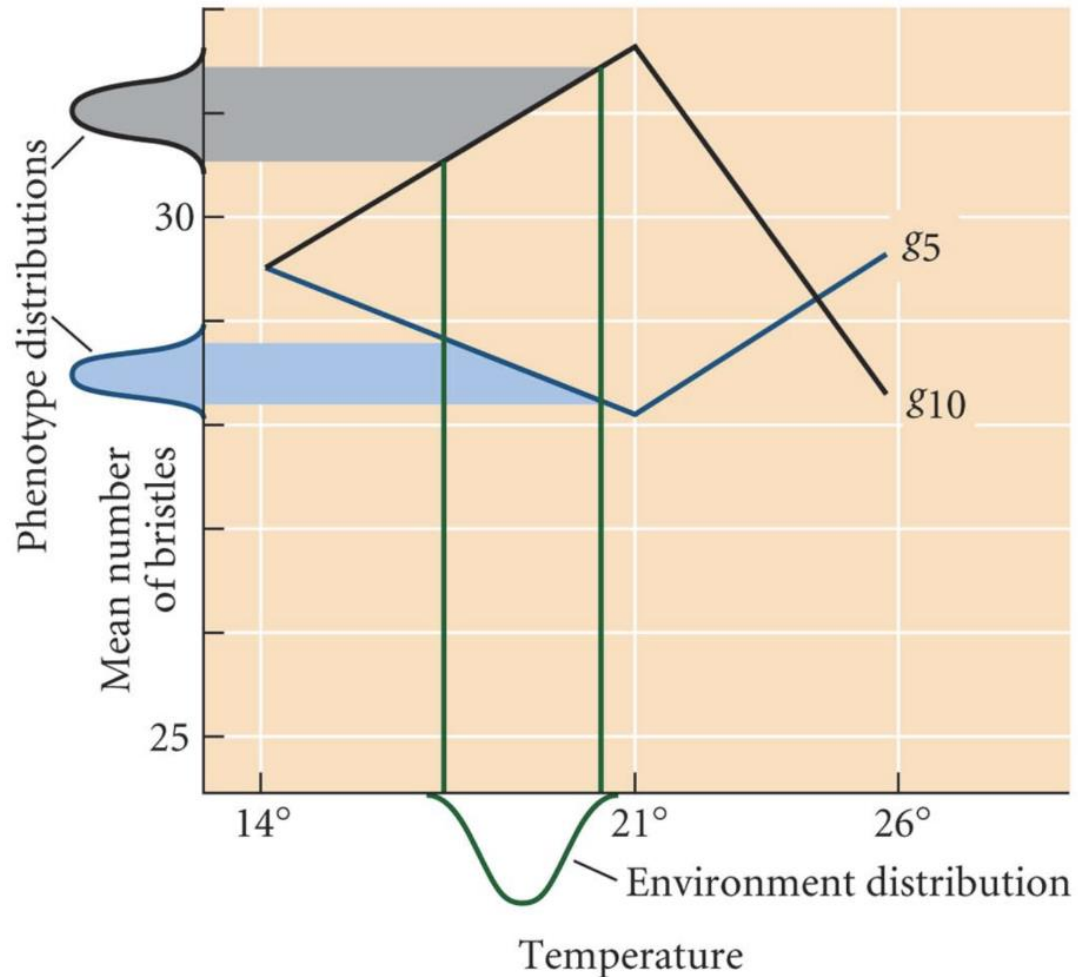


GxE interactions

- GxE for number of bristles produced by fruit flies
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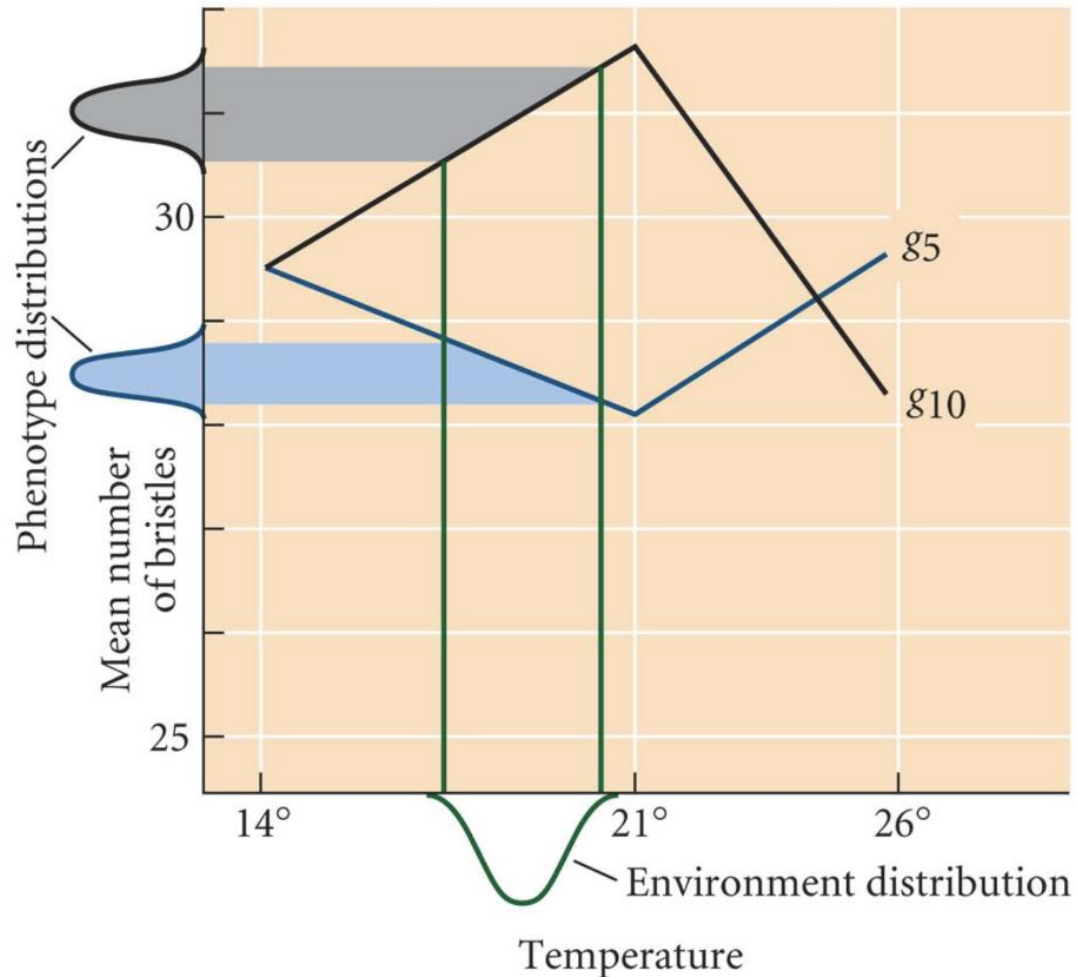


Amount of phenotypic variation due to genetic differences depends on temperature flies exposed to during development

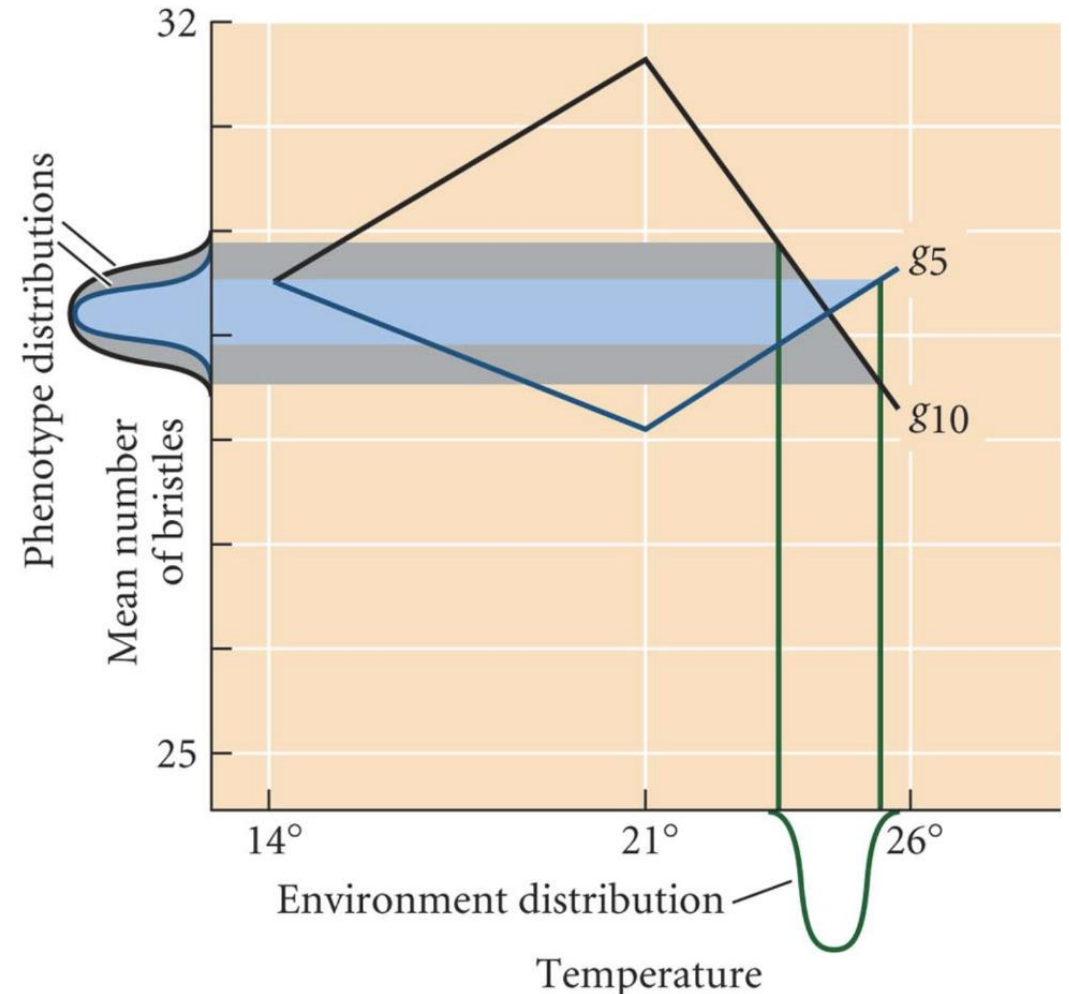


Reared around 20 degrees = more variation

Amount of phenotypic variation due to genetic differences depends on temperature flies exposed to during development



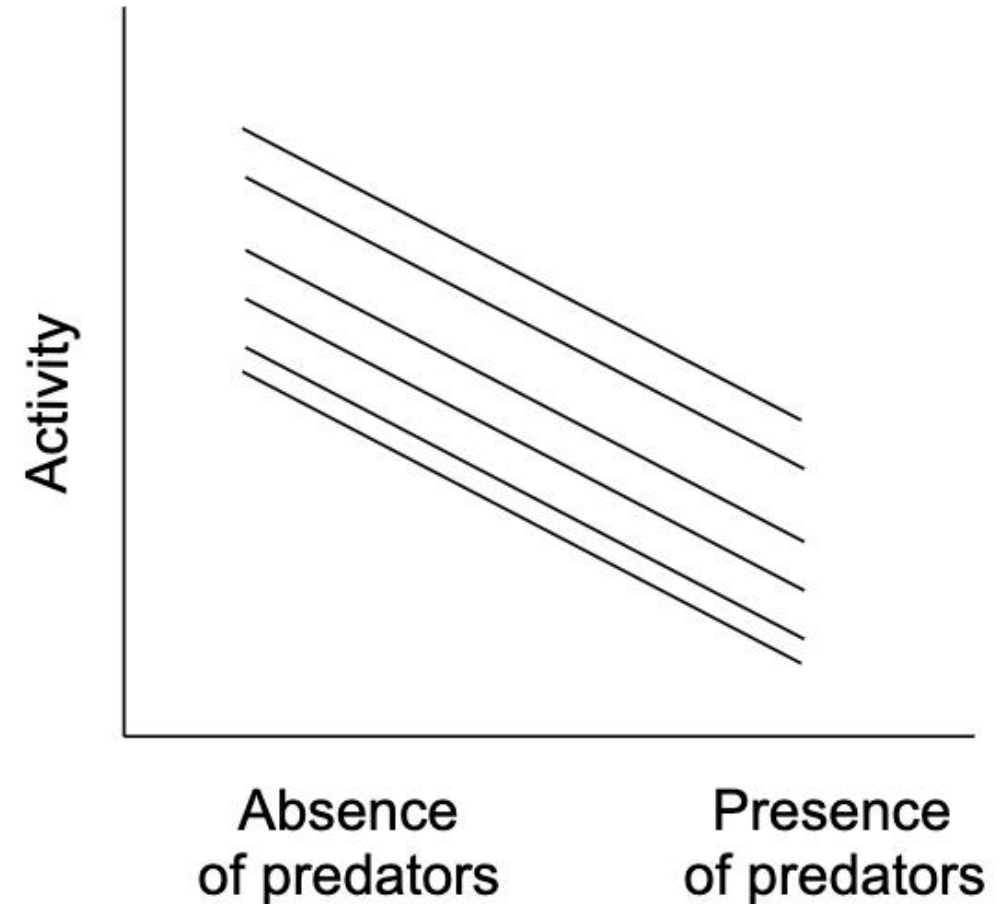
Reared around 20 degrees = more variation



Reared around 25 degrees = less variation

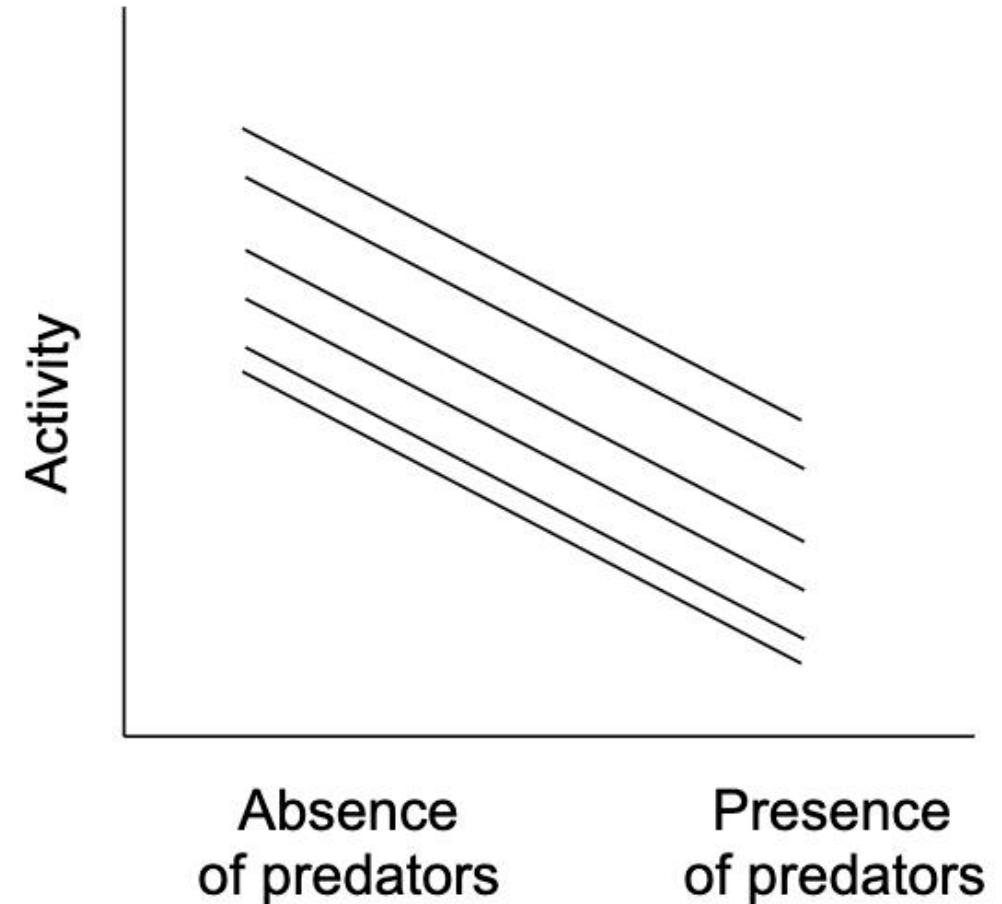
Behavioral reaction norms

- Lines represent individual animals rather than genotypes
 - Could still be genotype if you knew genotype linked to behavior (this can be difficult to find)



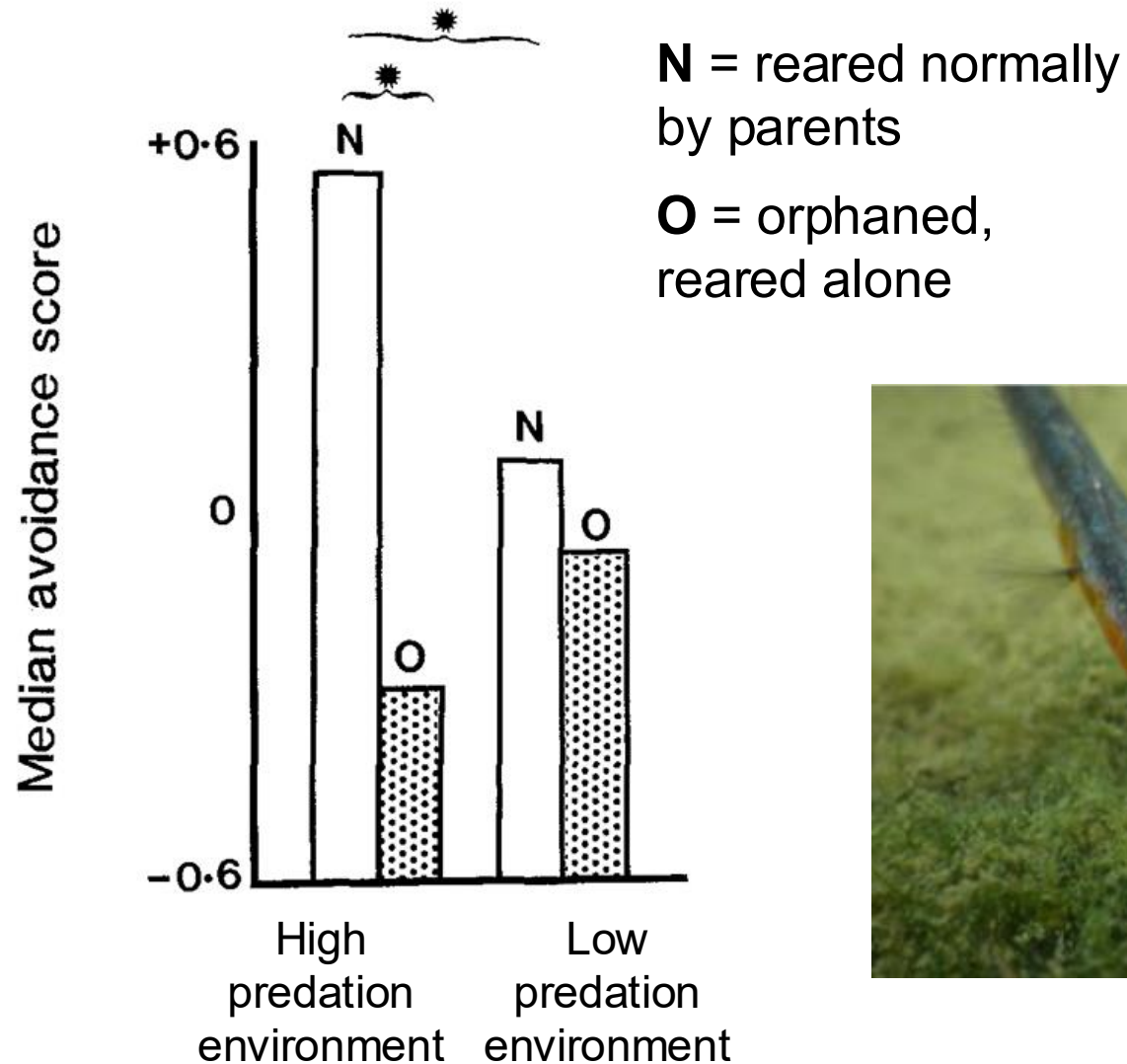
Behavioral reaction norms

- Lines represent individual animals rather than genotypes
 - Could still be genotype if you knew genotype linked to behavior (this can be difficult to find)
- Behavior is often more plastic than other phenotypes because it is often based on immediate responses to the environment



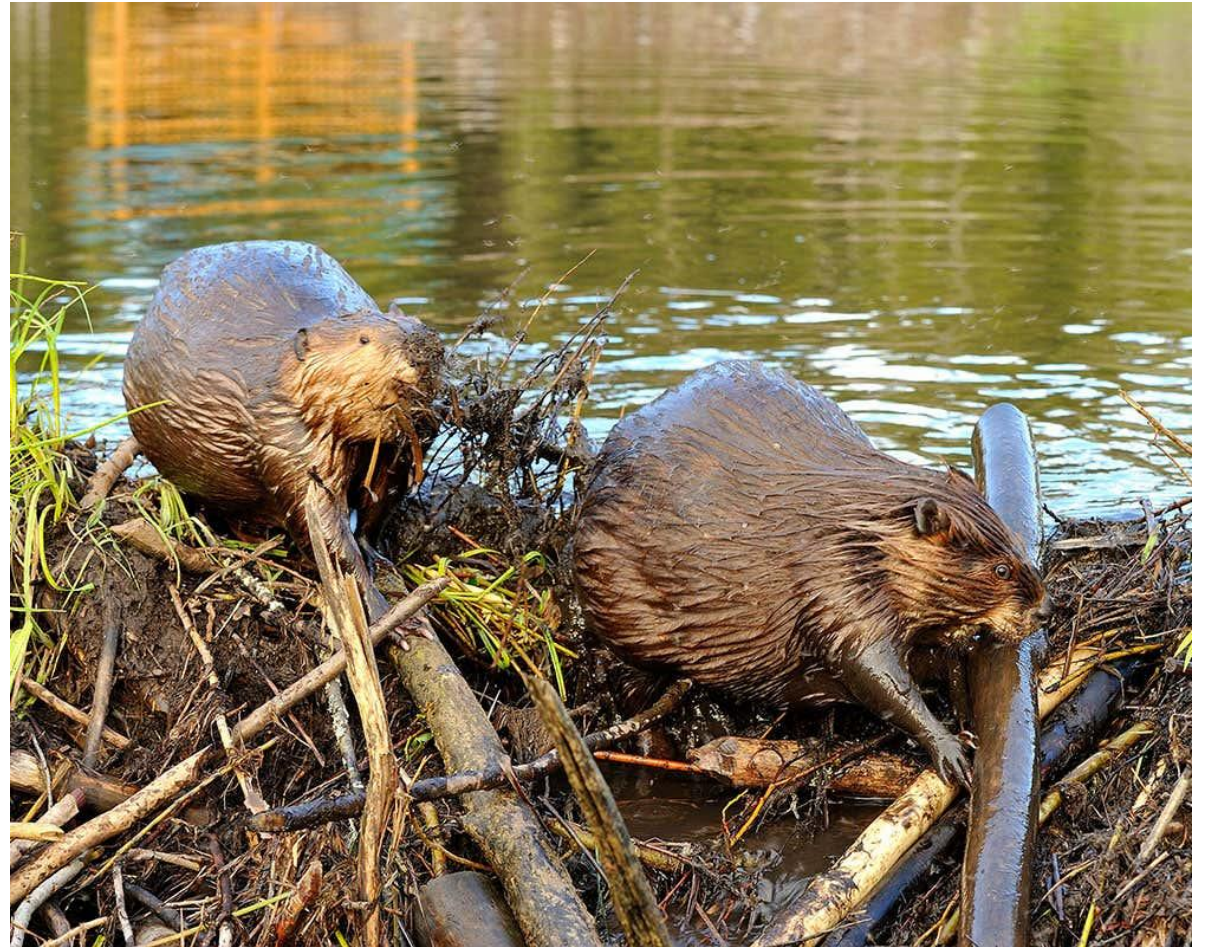
Behavioral reaction norms

- Interpret this figure. Is there a GxE / behavior x E interaction? Why or why not?



Animals can also alter their environment through behavior

- Beaver dams
- Termite mounds
- Earthworms moving around soil
- Social environment (i.e., selecting group to live in)
- Generates feedbacks between behavior and environment



Trans-generational plasticity

- Plasticity not just in individuals
 - Rat pups that had more attentive mothers (licking, grooming) had increased exploratory behavior



Weaver et al 2004

Trans-generational plasticity

- Plasticity not just in individuals
 - Rat pups that had more attentive mothers (licking, grooming) had increased exploratory behavior
 - **Aphids exposed to predation risk had reduced risk-taking behavior**
 - Epigenetic modifications



Keiser & Mondor 2013

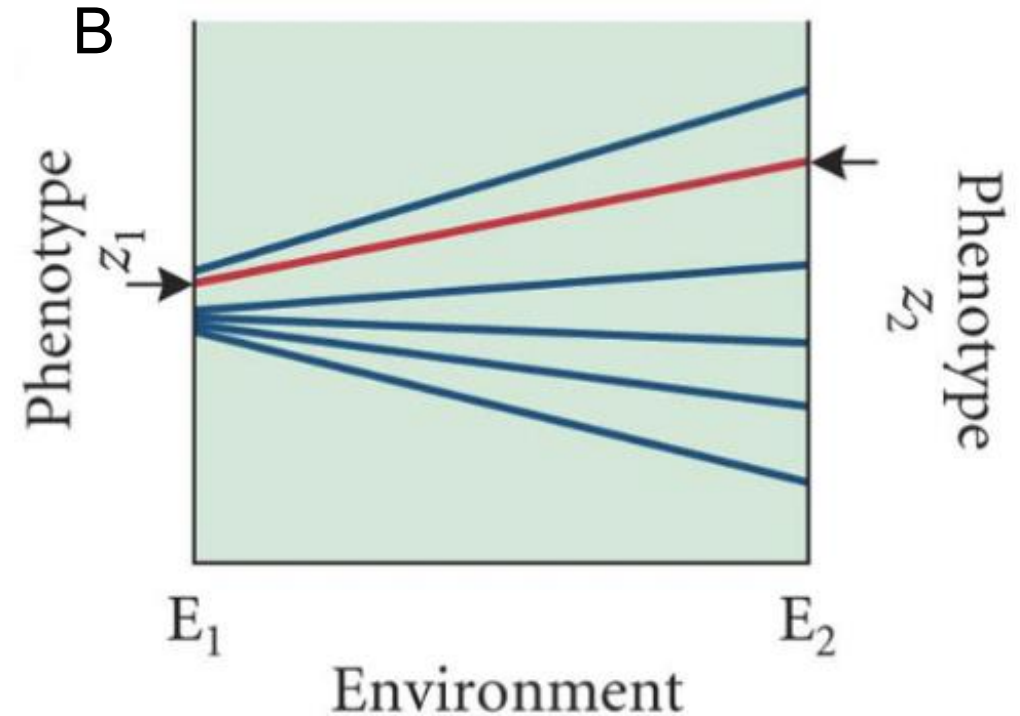
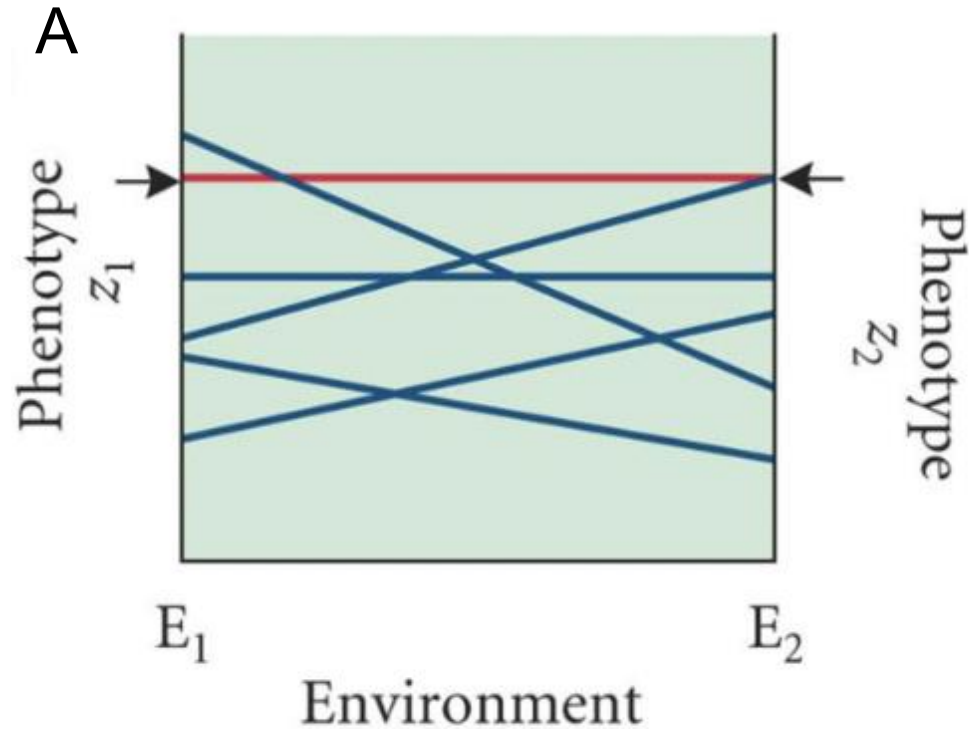
Evolution of plasticity

- Requires:
 - Genetic basis of plasticity
 - Variation in plasticity (some individuals more plastic than others)
 - Optimal phenotype (i.e., highest fitness) is different across environments



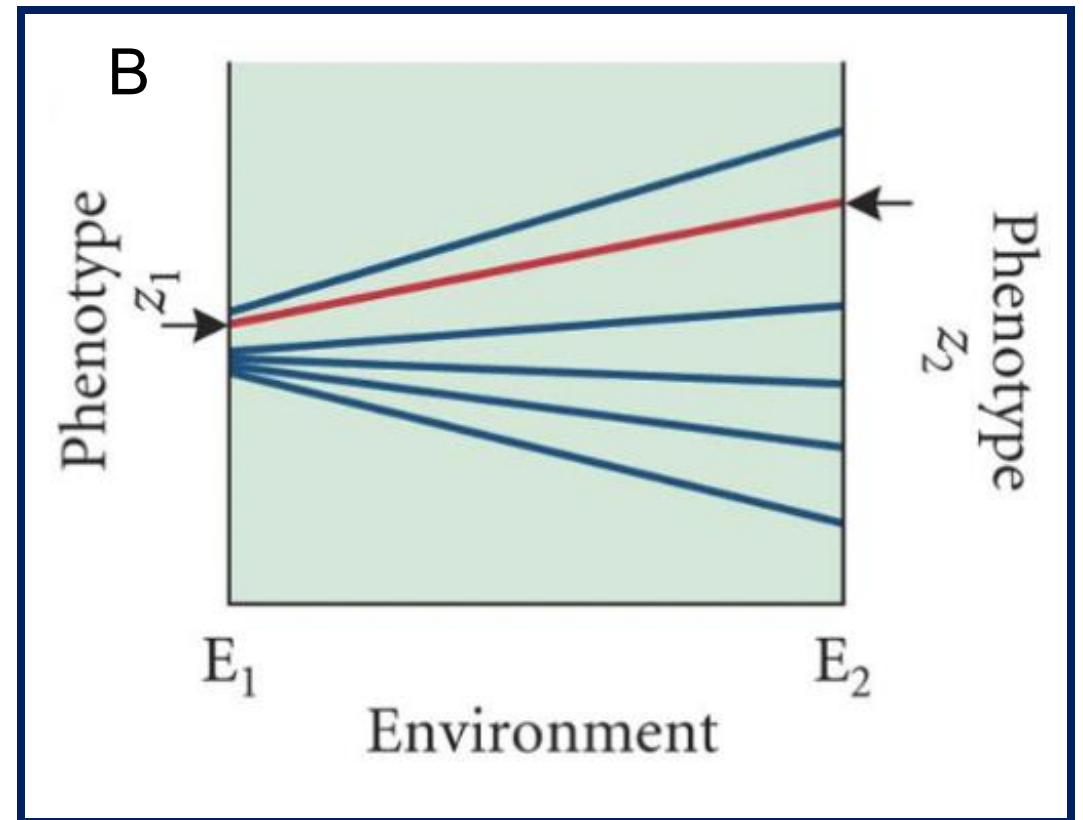
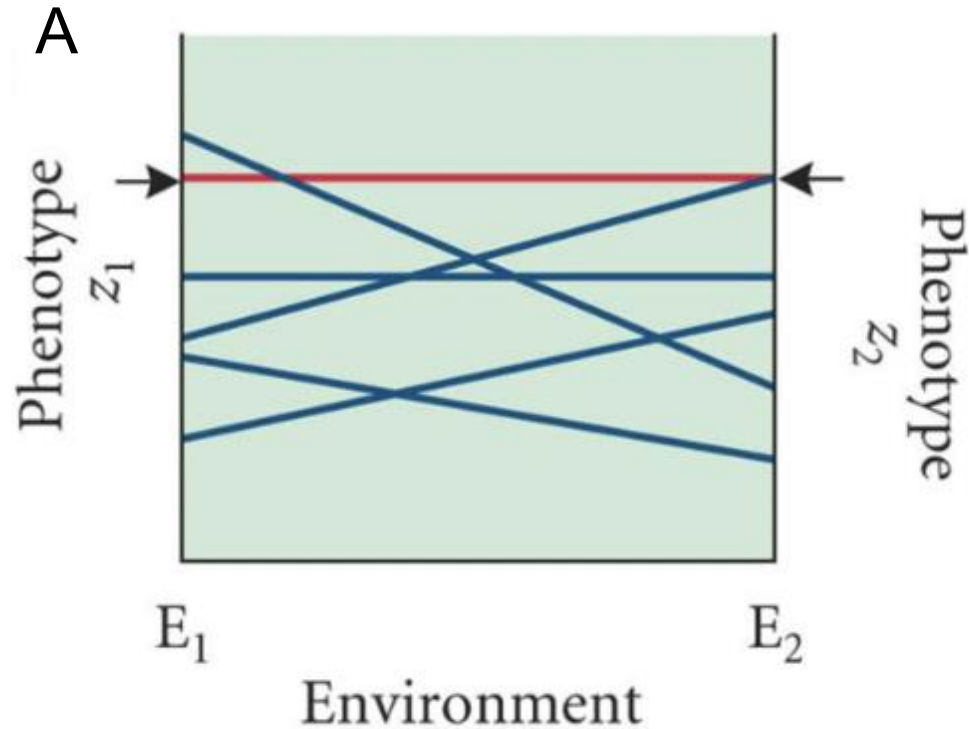
Evolution of plasticity

- In which scenario is plasticity more likely to evolve?
 - Red = optimal phenotype



Evolution of plasticity

- In which scenario is plasticity more likely to evolve?
 - Optimal phenotype differs across environments



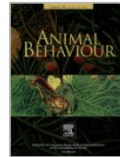
Evolution of plasticity

- Benefits of plasticity
 - Helps you live in multiple habitats
 - Can respond to disturbance / environmental changes
 - Expand resource pool
- Costs of plasticity
 - Energetic / physiological costs
 - Lack of specialization (jack of all trades, master of none)
- Limits to plasticity
 - Genetic or physiological constraints
 - Reliability of environmental cues (environment too unstable to keep up)



Animal Behaviour

Volume 140, June 2018, Pages 93-98



Colour plasticity alters thermoregulatory behaviour in *Battus philenor* caterpillars by modifying the cue received

Matthew E. Nielsen ^a  , Eran Levin ^b, Goggy Davidowitz ^b, Daniel R. Papaj ^a

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- Black caterpillars seek refuge from heat at lower temperatures than red caterpillars (dark coloration absorbs more light) – why?



- Black caterpillars seek refuge from heat at lower temperatures than red caterpillars (dark coloration absorbs more light) – why?
 - H1: Differences in the intensity of cues that generate behavior
 - H2: Differences in behavioral reaction norms



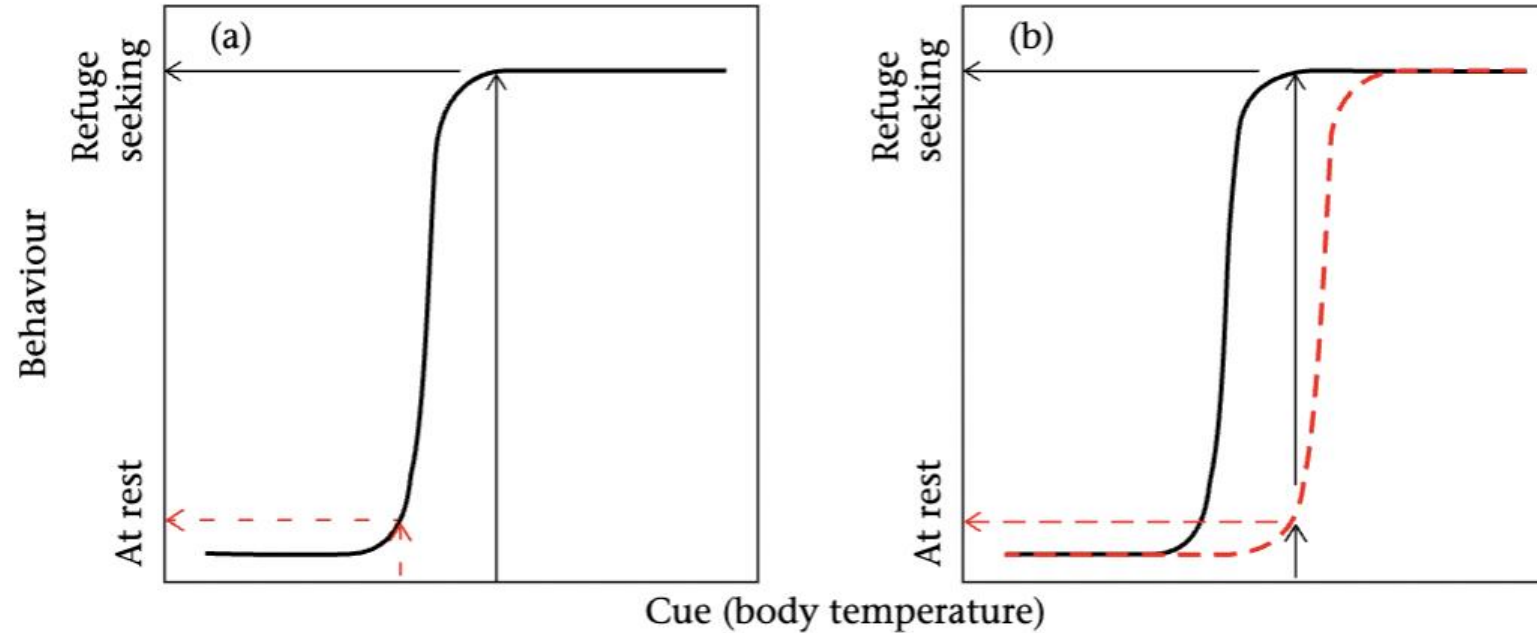
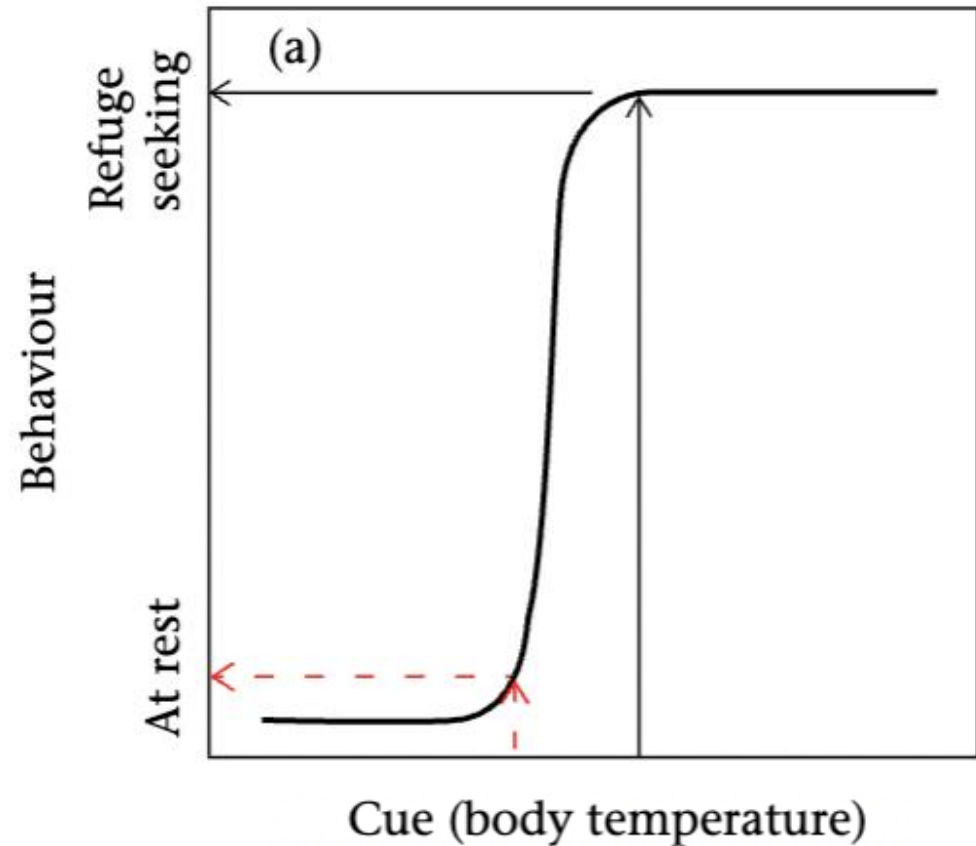
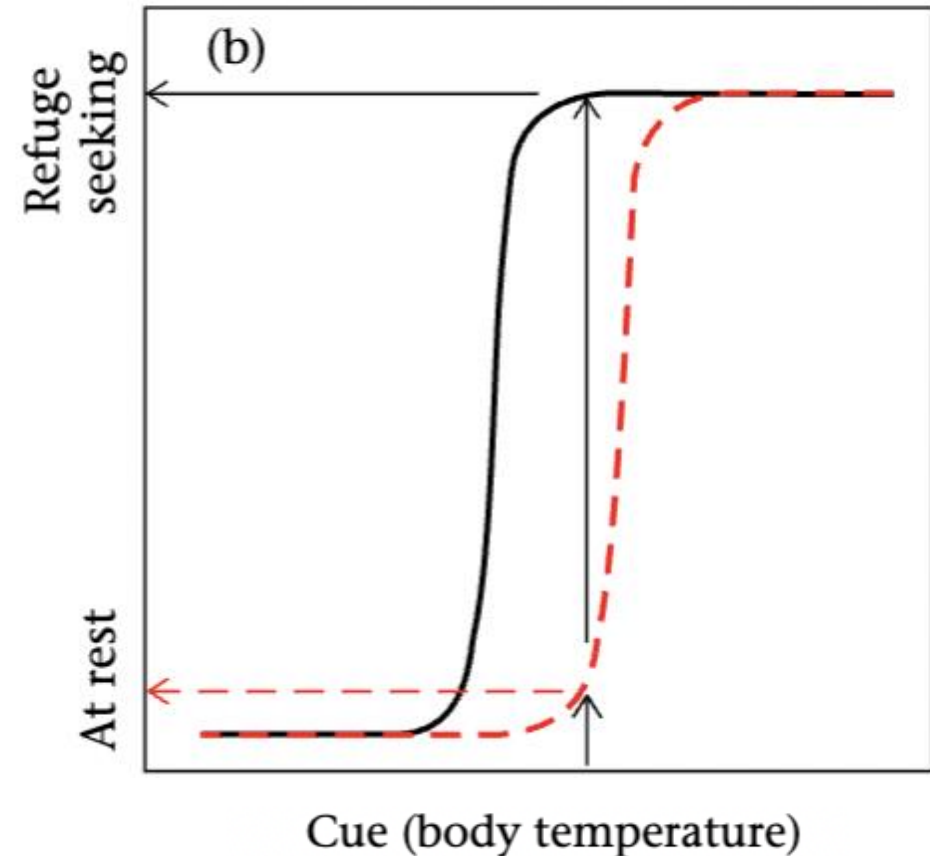


Figure 1. Hypothetical diagram of two mechanisms by which developmental plasticity could alter the expression of a behaviour, using colour plasticity and refuge-seeking behaviour in *Battus philenor* as an example. The X axis shows the value of the cue for the behaviour (body temperature for refuge seeking in *B. philenor*), and the Y axis shows the behavioural response, in this case modelled as simple on–off dichotomy with two discrete states (either at rest or refuge seeking). The curve represents the behaviour's reaction norm, whose key feature in this case is the temperature threshold at which the behaviour starts. Arrows are used to show the behavioural response associated with a particular

- H1: Differences in the intensity of cues that generate behavior
- Red and black caterpillars have the same temperature threshold for when to seek refuge from heat, but external morphology leads black caterpillars to warm up faster and move before red

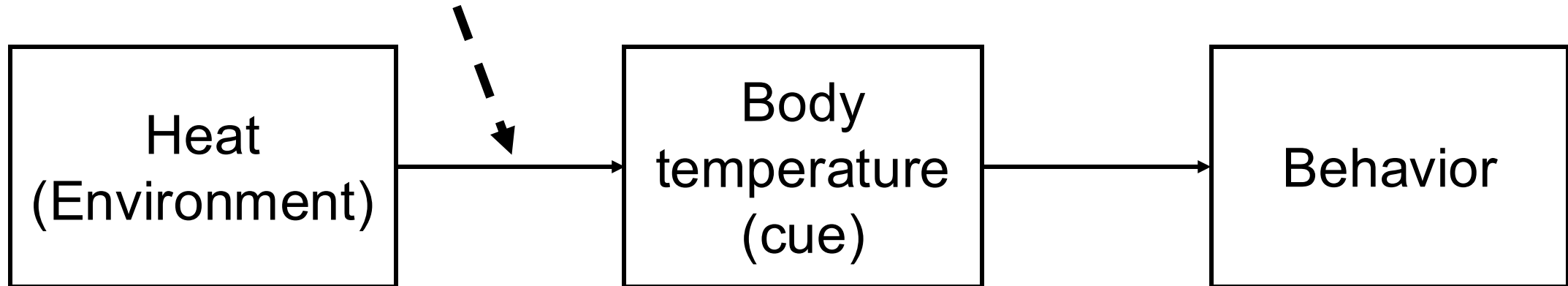


- H2: Differences in behavioral reaction norms
- Physiological, sensory and/or biochemical differences lead to different behavioral responses; i.e., the temperature at which you will seek refuge from heat



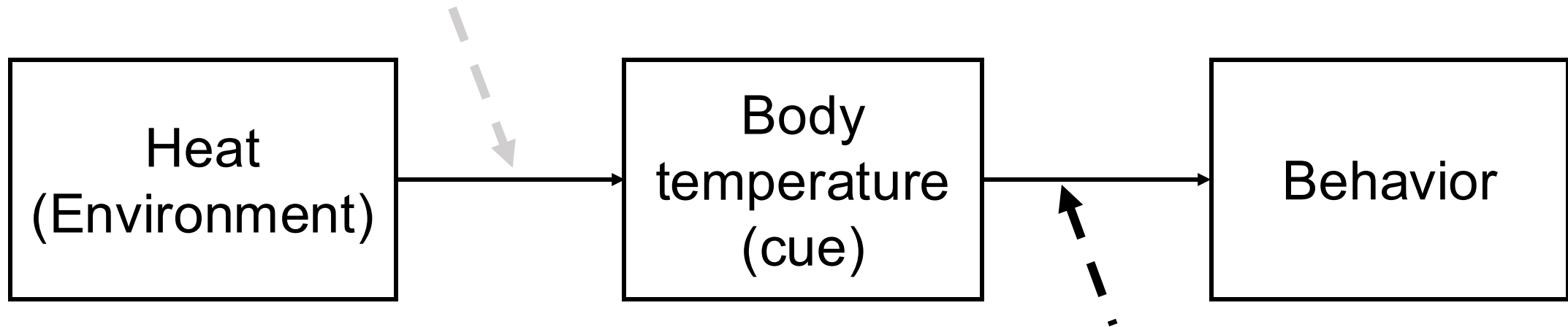
H1: Morphology shapes cue *before* it is processed

1. Heat from environment
2. Dark coloration makes it hotter (altering cue)
3. Animal processes this cue
4. Behavioral response



H1: Morphology shapes cue *before* it is processed

1. Heat from environment
2. Dark coloration makes it hotter (altering cue)
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H2: Physiology shapes reaction norm *after* cue is processed

1. Heat from environment
2. Animal processes this cue
3. Metabolic rate, hormones, other physiological processes differ across temperatures (altering reaction norm)
4. Behavioral response

- To test these hypotheses, measured use of refuge when exposed to radiant vs. conductive heat
- Conduction = heat transfer via direct contact
- Radiation = heat transfer via electromagnetic waves (e.g., light)



- To test these hypotheses, measured use of refuge when exposed to radiant vs. conductive heat
- What prediction would fit each hypothesis?



- To test these hypotheses, measured use of refuge when exposed to radiant vs. conductive heat
- What prediction would fit each hypothesis?
- H1 (cue intensity) = behavior should differ in red vs. black in radiant but not conductive heat condition
 - Radiant light makes it hotter for black = more intense cue



- To test these hypotheses, measured use of refuge when exposed to radiant vs. conductive heat
- What prediction would fit each hypothesis?
- H2 (reaction norm) = behavior should differ in both conditions
 - Regardless of where heat comes from, physiological response to heat differs

